

BurnTrack Project — Restructuring

June 25, 2026

1 What changed

All old, deprecated, or unused files have been moved to `archive/`. Nothing was deleted — everything is preserved and still accessible.

2 Archive layout

<code>archive/</code>	
<code>corrector_old/</code> (deprecated Jun 24)	<code>burntrack/corrector/</code> - pre-leakage-fix MLP corrector Replaced by <code>burntrack/correcteur final/</code> (31-feature MLP, clean)
<code>models_old/</code> <code>ensemble</code> , <code>atlas_v2</code> , <code>v3</code>	Old model checkpoints: RF, XGBoost, CatBoost, Current model: <code>burntrack/correcteur final/checkpoints/</code> <code>burntrack_mlp_minimal.pt</code>
<code>configs_old/</code>	<code>configs/</code> - old YAML/JSON config files
<code>scripts_old/</code> <code>validation scripts</code>	Old training, debug, synthetic generation, and Current scripts: <code>scripts/download_*.py</code> , <code>scripts/build_*.py</code>
<code>rapport_old/</code> <code>old report PDF</code>	Page-padding scripts (<code>inflate</code> , <code>simplify</code> , <code>stretch</code>) + Current report: <code>rapport/rapport_plbd_groupe7.pdf</code>
<code>docs_old/</code> <code>docs</code>	Old session logs, verification report, <code>pm.md</code> handoff
<code>data_old/</code> <code>key</code>	Old synthetic dataset generator, FIRMS zip, service Current data: <code>data/processed/</code> <code>south_africa_manual_dataset.csv</code> , <code>data/processed/table_mountain_2021.csv</code> , <code>data/processed/local_firms_ros_observed.csv</code>
<code>figures_old/</code>	Old figures (<code>metrics.json</code> , scatter plots)
<code>misc/</code> <code>_local/</code>	Prototype photos, Fritzing schematic, swap file,

3 Current active layout

<code>burntrack-simulation/</code> <code>burntrack/</code>	
<code>correcteur final/</code> <code>leakage)</code>	Current MLP corrector (31 features, clean, no
<code>data/</code> <code>py</code>	<code>firms.py</code> , <code>weather.py</code> , <code>real_dataset.py</code> , <code>burned_area.</code>
<code>engine/</code>	<code>rothermel.py</code> , <code>fuel_models.py</code>

utils/	logging.py, simulation.py
cellular_automaton/	Grid, rules, simulation, mlp_corrector
experiments/	validate_real_fire.py, calibrate_ca.py,
real_fire_scenarios.yaml	
rapport/	LaTeX report + figures + PDF
risk_map/	Bouskoura risk map generator
robot_nav/	D* Lite planner
robot_vision/	YOLO + CNN vision
scripts/	Download + build scripts (active only)
tests/	Unit tests
visualization/	WebSocket server + 3D UI
ReadIfAgent.md	Agent handoff doc (current state)
pm.md	Agent handoff prompt
pyproject.toml	Package config
requirements.txt	Dependencies
.env	API keys (gitignored)

4 Key files for the report

File	Purpose
rapport/rapport_plbd_groupe7.pdf	Final report PDF
rapport/rapport_plbd_groupe7.tex	Main LaTeX source
rapport/rapport_part2.tex	Part 2 (validation results, conclusion)
rapport/figures/	All report figures
experiments/out/	Validation outputs (JSON, figures, summary)
south africa data/	ERA5 GRIB, FIRMS shapefiles, SRTM, WorldCover, Landsat scenes

5 Validation metrics (final)

5.1 Knysna 2017 — Landsat-fused (dNBR threshold 0.05)

Metric	Value
IoU strict	0.516
IoU generous	0.620
F1 strict	0.681
F1 generous	0.765
Precision	0.793
Recall	0.597

5.2 Table Mountain 2021 — Landsat-fused (dNBR threshold 0.40)

Metric	Value
IoU strict	0.070
IoU generous	0.067
F1 strict	0.131
Precision	0.933
Recall	0.071
AUC	0.532

5.3 Table Mountain 2021 — FIRMS-only (original)

Metric	Value
IoU strict	0.02
IoU generous	0.05
F1 generous	0.10
AUC	0.55

6 How to run validation

```
# Knysna (fastest - ~54s with 5 workers)
venv/bin/python -u experiments/validate_real_fire.py --id knysna_2017_06

# Table Mountain (~2 min)
venv/bin/python -u experiments/validate_real_fire.py --id
    table_mountain_2021_04
```

7 How to rebuild the report

```
cd rapport && pdflatex -shell-escape rapport_plbd_groupe7.tex
```